Check for updates

Bradley Camburn[1]
Department of Mechanical Industrial and Manufacturing Engineering,
Oregon State University,
2000 SW Monroe Avenue,
Corvallis, OR 97331
e-mail: bradley.camburn@oregonstate.edu

Yuejun He
International Design Centre,
Singapore University of Technology and Design,
8 Somapah Road,
Singapore 487372
e-mails: yuejun_he@mymail.sutd.edu.sg;
yuejun_he@mymail.sutd.edu.sg

Sujithra Raviselvam
International Design Centre,
Singapore University of Technology and Design,
8 Somapah Road,
Singapore 487372
e-mail: sujithra_raviselvam@mymail.sutd.edu.sg

Jianxi Luo
International Design Centre,
Singapore University of Technology and Design,
8 Somapah Road,
Singapore 487372
e-mail: jianxi_luo@sutd.edu.sg

Kristin Wood
Fellow ASME
International Design Centre,
Singapore University of Technology and Design,
8 Somapah Road,
Singapore 487372
e-mail: kristinwoodcreo@gmail.com

Machine Learning-Based Design Concept Evaluation

In order to develop novel solutions for complex systems and in increasingly competitive markets, it may be advantageous to generate large numbers of design concepts and then to identify the most novel and valuable ideas. However, it can be difficult to process, review, and assess thousands of design concepts. Based on this need, we develop and demonstrate an automated method for design concept assessment. In the method, machine learning technologies are first applied to extract ontological data from design concepts. Then, a filtering strategy and quantitative metrics are introduced that enable creativity rating based on the ontological data. This method is tested empirically. Design concepts are crowd-generated for a variety of actual industry design problems/opportunities. Over 4000 design concepts were generated by humans for assessment. Empirical evaluation assesses: (1) correspondence of the automated ratings with human creativity ratings; (2) whether concepts selected using the method are highly scored by another set of crowd raters; and finally (3) if high scoring designs have a positive correlation or relationship to industrial technology development. The method provides a possible avenue to rate design concepts deterministically. A highlight is that a subset of designs selected automatically out of a large set of candidates was scored higher than a subset selected by humans when evaluated by a set of third-party raters. The results hint at bias in human design concept selection and encourage further study in this topic. [DOI: 10.1115/1.4045126]

Keywords: artificial intelligence, collaborative design, conceptual design, creativity and concept generation, design automation

1 Introduction

Design concept ideation often consists of multiple phases. Studies have demonstrated a correlation between the development of a large quantity of ideas and the identification of highly creative ideas [1,2]. Thus, one desirable approach to assist designers, design teams, and organizations is to create approaches that enable the generation of many ideas.

When deploying crowdsourcing to support early stage concept generation, preliminary concepts may number in the thousands [3]. Designers are often faced with a daunting task of categorizing these potential solutions, tagging keywords, and identifying the most innovative concepts to carry forward. To date, research in computational ideation demonstrates the use of graph topologies to automatically classify concepts using deep learning [4] or make new recommendations based on proximal solutions to a candidate solution in structured networks such as the patent database [5,6]. Previous theoretical work demonstrates mathematical foundations for calculating creativity scores from structured functional terms [7,8].

The study reported in this paper extends from these foundations and related work by the authors [3,9,10], which includes assessing morphological differences in large idea sets as a whole [3] and presenting an approach for the automated assessment of design concepts that are written in natural language without any pre-established meta-tagging (such as is present in the patent database). The method achieves automated assessment of creativity via quantitative metrics. Testing is applied to human generated concepts expressed in natural language. Early stage concept categorization and selection may feed into other generative design tools such as those for surrogate and evolutionary prototyping environments or even agent-based design [11–15] (see Table 10 of Appendix A for a definition of key terms).

Metrics influence the way decisions are made [16]. Design creativity metrics are used to quantify the results of an ideation session or to select candidate designs as part of a design sprint [17]. However, it takes training, repeatable coding methods, and significant human effort to evaluate these measures. Automated assessment of design metrics, such as *Novelty*—a measure of how unique a concept is relative to others [18]—and *Level of Detail*—a measure of how clearly a design is described [19]—could enable more rapid and repeatable workflow in early stages of design and design sprints.

The standard approach to concept assessment is executed through human expertize; therefore, results will vary depending on the evaluators and the various creativity metrics employed [7,8,18,20]. A successful implementation of more automated approaches would provide designers and the design research community with a benchmarking tool for repeatable assessment of design ideation outcome. Such an implementation likewise would allow for the integration of findings across multiple studies, organizations, or institutions.

This work explores one possible approach in utilizing machine learning (ML) and automating the assessment and sorting of design concepts. Two metrics are developed. They are analogous

[1]Corresponding author.

Contributed by the Design Theory and Methodology Committee of ASME for publication in the Journal of Mechanical Design. Manuscript received July 27, 2019; final manuscript received September 13, 2019; published online October 9, 2019. Assoc. Editor: Mark Fuge.

 Copyright © 2020 by ASME

from http://asmedigitalcollection.asme.org/mechanicaldesign/article-pdf/142/3/031113/6649246/md_142_3_031113.pdf?casa_token=JvwSKaiw1oIAAAAA:HtUkRLGCuC63WS_0PHGIPVY4zKH5xZbvnFwU8ToUCo_w9Zyd3l19WtAa2ZSjYLxf02k2w by Universita Di Pisa user on 25 Novembe

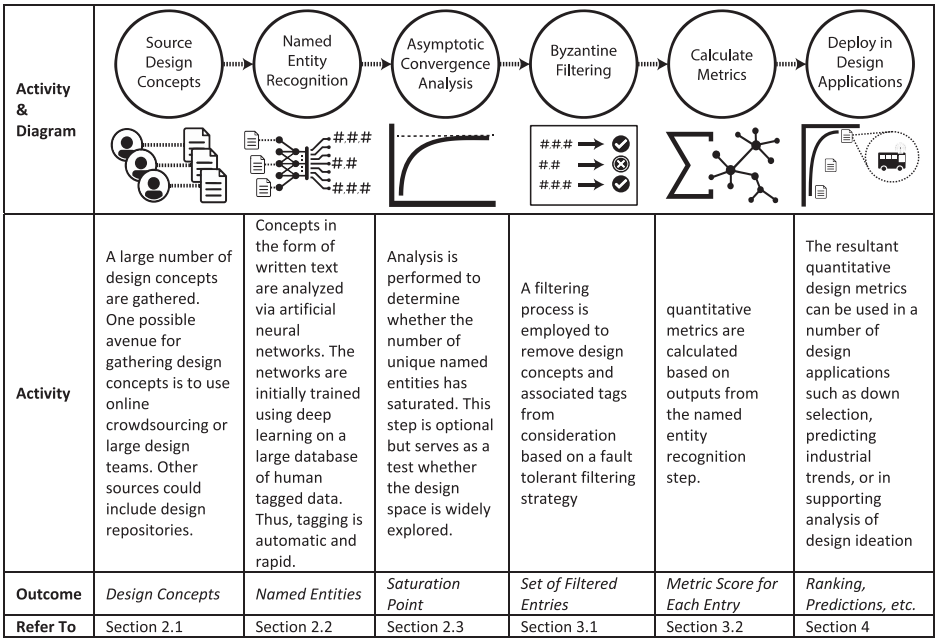


Fig. 1 Method architecture for ML-based concept evaluation. Standalone platforms are leveraged for crowdsourcing and ontological analysis. Aside from the proposed method, the strategy of Byzantine filtering and development of two ML-based metrics for concept selection are introduced in this work for the purpose of design concept selection.

to novelty and level of detail, respectively. Verification tests are applied to evaluate the strengths and weaknesses of the approach. Figure 1 illustrates our method for ML-based concept evaluation.

1.1 Research Approach. This work begins with an introduction of a method for measuring novelty and level of detail of a design concept automatically, and the subsequent experimental assessment of outcomes from this method. The method uses ML to provide a quantitative assessment of design concepts. The method architecture is outlined in Fig. 1. Each is later discussed in detail. Next, the study addresses three research questions (below) through empirical trials. The target objectives for our ML approach map to the fundamental purpose of evaluation through creativity metrics. Design metrics should provide repeatable assessment of ideation output to enable empirical studies as well as assist design practice. The three target objectives to design assessment, along with a corresponding research question to each objective, are as follows:

(1) *Rate design concepts*: How does the ML-based approach compare to human evaluation of design concepts, i.e., what is the inter-rater agreement between human and ML-metric assessment of design concepts for a sample set of designs? (see Sec. 4.1).
(2) *Identify novel designs*: Do creativity metrics implemented within the ML-based approach assist in the identification and selection of concepts that are novel from a large sample, and how do the resulting selections compare to time limited human selection of designs? (see Sec. 4.2).
(3) *Predict technology trends*: How do the ML-based selected designs compare to successfully implemented designs in industry, based on a comparison of selected concepts after a waiting period (of three years) from an initial ideation session? (see Sec. 4.3).

The paper details steps from the method (Fig. 1) in more detail. The performance of this process in evaluating design concepts is then tested in a series of controlled experiments. For ease of readability, the same case studies are used in introducing the method as well as to evaluate the method in controlled experimentation. In Sec. 2, an exemplar process used for crowdsourcing ideas and the ontological analysis are discussed. In Sec. 3, Byzantine filtering and ML-based creativity metrics are introduced and defined. In Sec. 4, empirical studies for each of the research questions are reported.

Table 1 Overview of the four opportunity-problem cases

Case #	Opportunity-problem statement cases	Source	Type
Case-1	"Design future public transportation in your city"	In-person crowdsourcing	Problem driven
Case-2	"Imagine a new transportation system in your neighborhood—Describe it!"	Distributed crowdsourcing	Problem driven
Case-3	"Explore possible inventions given a new technology: Global positioning system (GPS) that is accurate to less than 1 cm"	Distributed crowdsourcing	Technology push
Case-4	"Explore possible inventions given a new technology: Weather prediction that is 100% accurate"	Distributed crowdsourcing	Technology push

2 Data Collection Method

2.1 Crowdsourced Concept Generation. Large numbers of design concepts are gathered using both in-person (assembled design groups) and online (using Amazon's Mechanical Turk—Mturk [21,22]) crowdsourcing. The results of these ideation efforts are then used to evaluate the proposed method.

2.1.1 Problem Statements. Four ideation cases are employed to generate a large number of design concepts. Table 1 lists these cases. The objective is to explore differing types of design opportunities/problems by considering: (i) technology push versus opportunity-problem driven and (ii) ideation by designers in a group setting versus online, distributed crowdsourcing.

Data for the in-person crowdsourcing were collected at a large-scale ideation workshop with over 2000 mid-career

Downloaded from http://asmedigitalcollection.asme.org/mechanicaldesign/article-pdf/142/3/031113/6649246/md_142_3_031113.pdf?casa_token=JvwSKaiw1oIAAAAA:HtUkRLGCuC63WS_0PHGIPVY4zKH5xZbvnFwU8ToUCo_w9Zyd3i19WtAa2ZSjYLxf02k2w by Universita Di Pisa user on 25 Novembe



Fig. 2 Large-scale data collection event, case-1 of Table 1

professional designers and engineering participants (see Fig. 2) (*case-1* in Table 1). The data for the remaining cases were collected through Amazon's Mechanical Turk [3,21] with users given a design prompt and an open-ended text entry box to submit a single design solution to the prompt (the raw data are hosted and may be used with proper citation[2]).

2.2 Machine Learning-Based Ontological Analysis. This section reports key functions and design of the machine learning-based toolchain used to extract ontological data from crowdsourced design concepts. Machine learning enables quantitative measurement of ontological properties of input text. For this method, the ontological properties of interest are the topic of the sentence, what entities are mentioned, and a measure of how related these entities are to the topic. In this method, sentence based or "natural language" descriptions of design concepts are analyzed. Categories are important as elements for creating tags at the sentence level. A neural network is trained for this purpose by matching dependency parse trees to determine broader classifications. This process occurs at the sentence level [23].

In this approach, several ontological characteristics of a given concept string are extracted automatically. For this study, these characteristics are determined through the use of the Textrazor application programming interface (API) [23,24]. Therefore, additional training or supervision is not required in this implementation of the method, as shown in Fig. 1. The properties are (1) entities or *one of millions of disambiguated unique keywords*, (2) topics or *a unique entry in an evolving database of higher level categories* (e.g., the hierarchical topical categories of Wikipedia) are keywords within the concept or sentence, and (3) categories which are taken from *a standard hierarchical classification which is fixed* [23]. IPTC media codes are one such classification used for categories. They are an industrial standard [25] used in classifying media on the web and are a hierarchical classification meta-tagging system for broad knowledge topics [26]. Topics and categories are determined at the sentence level through Relevance scores, *which are the contextual relatedness of each entity in a given concept (sentence) to the concept's topic*. Details of the categorization procedure and insight or purpose for performing each of these categorizations are listed in Table 2 and in documentation for the Textrazor API [23,24,27–29]. An illustration of the functional mechanics of entity tagging is depicted in Fig. 3.

Textrazor utilizes the FreeBase knowledge graph and other open source databases like Wikipedia as the core for determining relevance and training its neural networks to tag entities. FreeBase provides a structured graph across general domains of human knowledge. Other structured databases, such as patent databases, are limited to submitted and technical-system innovations. Innovative design spans broad opportunity spaces across products, services, systems, or even social innovations. Relationships across general knowledge are important as designers often develop solutions across many domains [30–32]. Some domains, such as service design, may be less represented in technical databases. Thus, our requirement is for a broad structured database which the FreeBase technology fulfills [33]. Lastly, this database should be consistently available to researchers and designers to enable repeatability of results. Archival copies of the FreeBase graph are available for free download [34], and it is a core structure for many influential ML-based technologies.

With the functions and design of the ML-based method now described, we describe how concepts are processed. First, an input string, i.e., an expressed concept, is tokenized, followed by spellchecking, word stemming, and detection of part of speech. These are common procedures available in most language processing toolkits.

Some works in natural language processing use entities but do not evaluate sentence level topics. This approach is sometimes called a "bag-of-words." Then, the similarity of two word bags can be compared using tools like Word2Vec [35]. However, while similarities are computed [3,4], the syntactic dependencies in a natural language concept are lost. For example, a "bicycle designed to act as a modular passenger train" is not the same as a "train designed with modules for bicycles and passengers," yet they have the same lemma ("bicycle," "train," "passenger," "carry," and "module"). Bag-of-word approaches have demonstrated substantial merit; however, they cannot capture concepts

[2]https://tinyurl.com/ML-metrics-IDC

Downloaded from http://asmedigitalcollection.asme.org/mechanicaldesign/article-pdf/142/3/031113/6649246/md_142_3_031113.pdf?casa_token=JvwSKaiw1oIAAAAA:HtUkRLGCuCc63WS_0PHGIPVY4zKH5xZbvnFwU8ToUCo_w9Zyd3119WtAa2ZSjYLxf02k2w by Universita Di Pisa user on 25 Novembe

Table 2 Tagging types extracted

Tag	Motivation	Method of extraction	ML training set
Entity	• Identify keywords • Disambiguate the correct sense	• Probability of co-occurrence • Determined by Word2vec similarity score and Part of Speech tag, via syntactic dependency, part of speech (P.O.S.) tagging trained with recursive neural networks	• DBpedia, Wikipedia, Newswire (unique ID number)
Topic	• Identify the overarching sentence topic • Employ evolving classifiers	• Machine learning is applied to identify patterns between combinations of specific entities and hierarchical topics within the Wikipedia database	• Wikipedia (hierarchical evolving categories)
Category	• High level categorization of subject • Employ fixed, hierarchical, classifiers	• Machine learning is applied to identify patterns between combinations of specific entities and the IPTC media codes	• Freebase [26], Wikipedia (Standardized IPTC media codes [27])
Relevance	• Measure of distance between a *Core Entity* and other *Entities* in a tagging effort	• Determined using Textrazor's Knowledge Graph which is an analogue to the Freebase knowledge graph, provides a 'distance' between any two unique entities	• Freebase

Note: Each tag has a unique associated neural network trained for detection of that type of tag. Each tag is identified separately using a dedicated neural network [16].

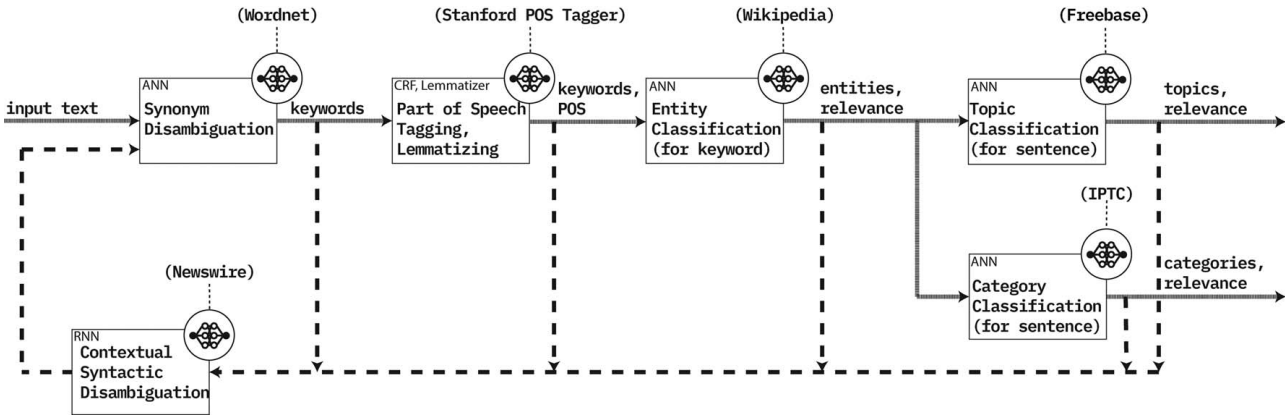


Fig. 3 Functional diagram for named entity recognition, high-level architecture. Key training sets and data structure flows are indicated where relevant. Use of artificial neural networks [54], conditional random fields [55], or recurrent neural networks [56] is indicated in the top left of each respective function block. At a high level, this structure enables entity classification to be parallelized with the topic and categorization to improve quality; this is primarily achieved through the use of a recurrent neural network for syntactic disambiguation (confirming subject-object and other sentence level hierarchical relationships).

that rely on the order, hierarchy, or sense of terms. The difference is anecdotally analogous to that between a pile of gears and a watch.

To classify unique entities, and their contextual dependency at the sentence level, a dependency parse tree is established for the lemma of each term. This parse tree establishes part of speech and, through association of subject and object, enables training of a recurrent neural network to infer, probabilistically, the sense of a word (see Fig. 3). For instance, dependency parsing enables a trained hierarchy to differentiate between "Ford" the motor car company and "ford" a shallow segment of a river through contextual information, e.g., the sentence also contains "motor" in a given clause. These tagging requirements were met by leveraging the Textrazor platform [23], where in-depth supporting literature includes [24,27–29,36].

2.3 Asymptotic Convergence Analysis. One of the potential benefits of crowdsourcing and automated concept evaluation is that it would support rapid evaluation of a large number of design concepts. However, it is unclear how many distinct design concepts may be present for a design space and whether there are in fact a sufficiently large number that automation is justified. Therefore, an asymptotic convergence analysis, using the algorithm outlined by Lim et al. [3], is employed. In this case, once a unique *Core Entity* is detected, it is not counted again if a new concept utilizes the same *Core Entity*. This test establishes first that there are many different categories of solutions and second that the data set is converging in terms of unique categories. There could be many more unique ideas when secondary entities are also considered. At approximately 1000 entries each, all four cases were saturated according to the criteria (from Ref. [3]) in each of the sampling sequences. Random sampling trials were employed to verify that this result is not an artefact of the sampling sequence. Thus, 1000 entries are used in testing.

Note that asymptotic convergence analysis was performed on results from *case-1* and *case-2*, from Table 1, in a previous study [3]. In the previous study, saturation was performed by human raters [3]. The human results indicated results from *case-1* (see Table 1 and Fig. 4) as saturating first. In Fig. 4, it can clearly be seen that, for *Entities*, the online results saturate first when saturation is implemented using *Core Entity*. The results were also tested for the tag of a *Category* using saturation analysis. In both analyses, the *Category* tag was saturated earlier (with 14 categories at approximately $n = 100$ concepts tagged) which was consistent with previous publications [3]. One possible explanation for this result is that humans may form bins in a manner more like the *Category* tagging approach. However, *Core Entity-based saturation* is used herein due to the more granular classification that they afford.

3 Design Concept Evaluation Strategy

3.1 Byzantine Fault Tolerant Filtering. When design concepts are gathered from a repository or crowdsourcing method,

oaded from http://asmedigitalcollection.asme.org/mechanicaldesign/article-pdf/142/3/031113/6649246/md_142_3_031113.pdf?casa_token=JvwSKaiw1oIAAAAA:HtUkRLGCuCc63WS_0PHGIPVY4zKH5xZbvnFwU8ToUCo_w9Zyd3119WtAa2ZSjYLxf02k2w by Universita Di Pisa user on 25 Novembe

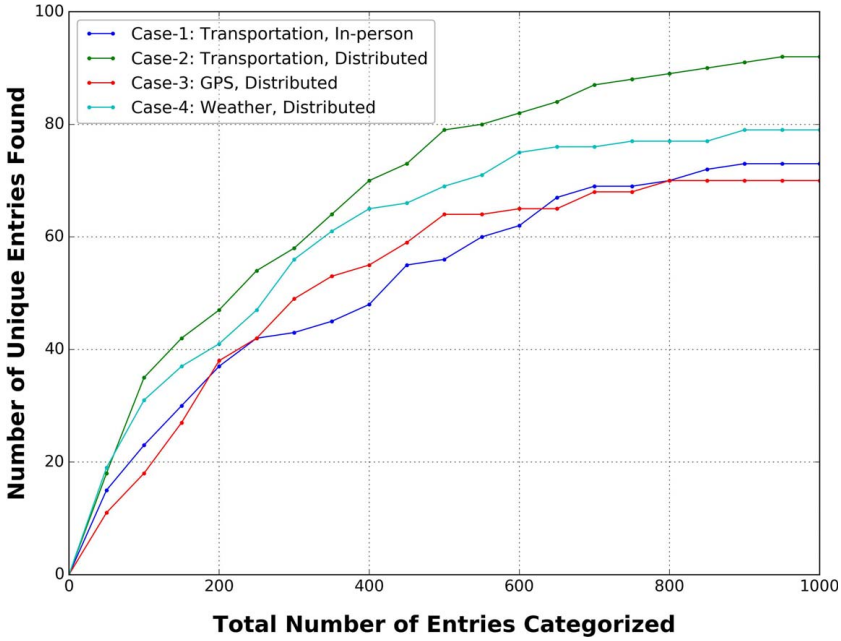


Fig. 4 Saturation of the *Core Entity* tagged as more concepts are evaluated. Note that each case saturated at or before 1000 entries.

Table 3 Example-based demonstration of the Byzantine fault tolerant filtering strategy

Sample concept	Tag type	Result	Filtering result	Notes
"The second new type of teleportation that I can think of is a flying car. It would either have wings or a hover pad on it. It would be able to go to landing zones and be able to go to places faster than what is normally possible"	*Entity* (first listed) *Topic* (first listed) *Category* (first listed)	Teleportation Aircraft (N/A)	*Filtered,* no *Category* tag was detected	*It is unclear whether this is a solution about teleportation or flying cars*
"Since rail is very expensive, I'd like to see more express bus routes, hopefully with their own lanes to keep them flowing faster than traffic and give an incentive to use them. Make the buses as comfortable as possible, again as an incentive for use"	*Entity* (first listed) *Topic* (first listed) *Category* (first listed)	Bus Vehicles Economy, business, and Finance > economic sector > transport	*Passed,* all three tags successfully detected	*The solution is about encouraging increased utilization of buses through an incentive scheme and roadway design*

there is no guarantee that all concepts will be well written or comprehensible. In a typical design process, a human evaluator can intuitively push these ideas aside. However, it can be a challenge to identify such a faulty entry automatically. This is because an incomprehensible entry might still be tagged by one of the three unique neural networks used in tag extraction (see Table 2 and Fig. 3). To reduce noise, concepts which were not successfully tagged at all three levels are removed. Failure to be categorized by one of the networks indicates that the concept may have been incoherent or poorly structured. This concept is adapted from fault tolerant control strategies. It may be referred to as voting-based fault tolerance or "byzantine" fault tolerance [37]. Other works have explored the use of multiple artificial neural networks to improve accuracy of predictions [38]. In principle, the theory is that if multiple distinct networks agree on a conclusion (in this case whether to filter an entry or not), this conclusion can be considered more reliable. A contextual demonstration of this approach, using a negative and positive example from the actual data, is presented in Table 3. Table 3 also provides examples of the output for entity, topic, and category tagging results.

Any accepted entry should be coherent. If this was not the case, ill-formed entries might offset the results, and the concept would be filtered. Ambiguous concepts are stringently filtered by a Byzantine fault strategy (see Table 3). It is critical that incoherent concepts are filtered as they might skew other analyses. Once the entries are tagged and filtered, ML-metric scores are evaluated.

3.2 Machine Learning-Metrics: Definition. For a design to be valuable, it should meet certain criteria. Ideation metrics should enable the assessment of whether a design meets the criteria for value. A design that is novel and well described to implement may be valuable. Thus, we propose metrics corresponding to evaluating these characteristics in design ideation data.

Recent research identifies a feature of novel designs as being drawn from analogies to another domain [31] or the synthesis of common and uncommon combinations [9].

In the tagging process, each entity is assigned a relevance score to the *Core Entity*. The *Core Entity* is determined through probabilistic mapping of entities and syntactic dependency using the Textrazor [23] knowledge graph. The relevance of each entity to the central topic is reported based on the network distance, or path length [39], between two given topics in the FreeBase knowledge graph [40], as determined by the ML analysis provided by Textrazor. Thus, the proposed metric to be used as a proxy for novelty at the concept level is "Span." *Span* is taken herein to be *the sum of distances of each entity in the concept to the central named Entity, as measured by minimum*

Downloaded from http://asmedigitalcollection.asme.org/mechanicaldesign/article-pdf/142/3/031113/6649246/md_142_3_031113.pdf?casa_token=JvwSKaiw1oIAAAAA:HtUkRLGCuCc63WS_0PHGIPVY4zKH5xZbvnFwU8ToUCo_w9Zyd3i19WtAa2ZSjYLxf02k2w by Universita Di Pisa user on 25 Novembe

spanning path within the knowledge graph. This aligns with standard novelty measures [7,9]. In a simple manner, distance in a network is the number of edges (steps) between two vertices (nodes or keywords). *Span* is calculated for a single concept as follows:

$$Span = \sum_{i=1}^{n} (1 - R_i) \tag{1}$$

where i is an individual *Entity*, n is the number of entities in the solution, and R_i is the relevance score for each individual *Entity*. For instance, if there were three entities in the concept with relevance score 0.5 and one with relevance score 0.1, the *Span* score would be $(1-0.1)\times 1 + (1-0.5)\times 3$.

Highly detailed design concepts are considered to be easier to interpret [41] and therefore likely to be easier to use in design moving forward. Although it does require more effort to create such designs [42], in this case, effort of creation is a sunk cost. Therefore, it adds value to identify detailed design concepts that are present in the results.

The proposed metric for the level of detail [42] is simply "Detail." *Detail* is taken herein to be *a basic count of named entities within the concept*. Note that a key precursor in applicability of this metric is the fact that the concept must be recognizable as pertaining to a distinct high-level *Category* due to the fault tolerant filtering strategy (see Table 3). The sum of entities is simply executed as follows:

$$Detail = n \tag{2}$$

where n is the number of entities in the solution and where this is a simple entity count.

A design, if useful, should be feasible to implement. Feasibility is often defined as the ability of a design to meet requirements. In some cases, the design requirements may not be strictly embedded in the design problem-opportunity statement, and automated evaluation of such requirements has proven challenging. However, some solution avenues may be recognizable by the designer as more likely to be feasible.

Leveraging this insight, a strategy is provided to assist designers in rapid down selection. Each design concept is tagged with multiple entities; one of these is defined as the *Core Entity* or entity with the highest relevance to the *Topic*. In other words, this Entity is most representative of the idea while being more specific than a Topic. To identify the *Core Entity*, simply extract the *Core Entity* with the highest Relevance to the *Topic* of the design concept.

$$R_{\text{core_entity}} = \max(R_i, R_n) \rightarrow Core_Entity \tag{3}$$

where $R_{\text{core_entity}}$ is the relevance of the *Core Entity*. The relevance of the *Core Entity* is the maximum relevance score of any entity in the solution to the Topic of the concept.

Taken together, these metrics provide a way of estimating how unique is an design concept (*Span*), how well it is described (*Detail*), and a guide for selecting a desirable solution category (*Core Entity*). A contextualized example follows in Table 4.

4 Experimental Assessment and Applications

Several tests are executed to address the research questions (Sec. 1.1) and evaluate the method. Testing includes ideation data from the four cases (see Table 1). Human inter-rater testing of novelty and level of detail are reported for a sample set of ideas. The human raters then reconciled to develop a final score. Inter-rater agreement of the final human scores and ML metrics are reported. The metrics are used to select high scoring ideas from the entire set (experimental). These are rated against other selection procedures which are ideas selected in a controlled time period by a third human (standard), random concept selection (baseline), and source text from an historical novel (control). The second test is to determine which selection approach was dominant. This test indicates whether the metrics can be "forward applied" and serve to identify ideas from a massive concept set. Finally, a comparison is made between the ML-based scores as a predictor of technology emergence in industry. This last study aims to determine if there is any potential connection between the results and innovations in industry. Details of each test are described below.

Table 4 Example design concept and metric calculation

Entry	"Charging station for electric cars that also has lots of electronic device chargers"	
Entities; relevance	1. Charging station; 0.3151	*Core Entity* = "Charging station"
	2. Electric car; 0.1534	*Novelty* = (1 − 0.3151) + (1 − 0.1534) + (1 − 0.09)
	3. Electronic device; 0.09151	*Detail* = 3

4.1 Rating Design Concepts: Inter-Rater Agreement.

One key opportunity for a ML-based design concept evaluation is to standardize experimental assessment of design concept outcomes in ideation studies. Inter-rater testing was applied to a sample set of 50 randomly selected concepts from each of the four cases, as shown in Table 1. Pearson's inter-rater agreement was used to determine the level of agreement between raters. The process included discussion of a small sample set, review, and agreement on the rubric, followed by the actual independent rating. Once ratings were complete, the raters discussed any discrepancies between raters to reach full agreement on each single rating. The objective for following this process is to increase the assessment quality. The rubrics for evaluating novelty and level of detail are shown in Tables 5 and 6,

Table 5 Metric scale used by the raters in assigning novelty values to each individual solution

Rank	Description
1	Solution exists and already serves the same in domain purpose, "everyday solution"
2	The technology solution combination exists in domain but is uncommon not "everyday"
3	Single new feature (new = approach or technology from another domain)
4	Two new features
5	Three or more new features together

Table 6 Metric scale used by the raters in assigning the level of detail

Rank	Description
1	Incomprehensible or no features are well explained
2	One feature is described
3	Two features are described
…	…
n	n features are described

Table 7 Human-to-human inter-rater Pearson's correlation agreement scores

	Ideation condition (see Table 1)			
	(1) Transport—in-person	(2) Transport—distributed	(3) GPS—distributed	(4) Weather—distributed
Novelty	0.98	0.84	0.74	0.74
Level of detail	0.91	0.88	0.86	0.73

Downloaded from http://asmedigitalcollection.asme.org/mechanicaldesign/article-pdf/142/3/031113/6649246/md_142_3_031113.pdf?casa_token=JvwSKaiw1oIAAAAA:HtUkRLGCuCc63WS_0PHGIPVY4zKH5xZbvnFwU8ToUCo_w9Zyd3119WtAa2ZSjYLxf02k2w by Universita Di Pisa user on 25 Novembe

Table 8 Human to ML-based inter-rater Pearson's correlation agreement scores

	Ideation condition (see Table 1)			
	(1) Transport—in-person	(2) Transport—distributed	(3) GPS—distributed	(4) Weather—distributed
Novelty/*Span*	0.39	0.41	0.36	0.47
Level of detail/*Detail*	0.51	0.55	0.21	0.69

respectively. Two of the authors performed rating, one was not aware of the research objective, hypotheses, or data sets until after rating was completed. The inter-rater agreement scores for human-to-human assessment are shown in Table 7.

The agreement between human and ML-based metrics is shown in Table 8. While the Pearson's correlation values are not as high as human-to-human scoring, there is a reasonable level of agreement. In addition, it may be noteworthy that this is a completely unsupervised approach, trained on generic knowledge graphs and thus broadly applicable. Second, the input data are fed as natural language, the interpretation of which can be ambiguous, or result in "multiple variant yet correct assessments" even between experts.

4.2 Identifying Novel Designs: Dominance Matrix. A second potential application of ML-metric-based design concept evaluation is to determine the most novel and well-described design concepts in a solution set. Thus, a selection exercise was conducted. The objective was to select ten highly novel, detailed designs to carry forward out of 1000 candidate designs.

The value ten was selected as it is considered the upper limit of working memory [43], and thus it is a reasonable value to assume for further evaluation in a design project. These concepts should be both novel and detailed. Thus, the exercise was for the given design opportunity-problem (case) and a set of 1000 candidate designs to "identify the ten (10) highest-scored, most novel and detailed designs in the original data set."

Concepts were selected using four competing strategies. In the first strategy, a human was given instructions to select the ten most novel and well detailed concepts in 12 min. They were given access to the entire set of 1000 concepts to review. First, the timing was based on a study of workplace behavior that a typical uninterrupted unit of work lasts approximately 12 min [44]. A professional was chosen to select designs; however, the solutions include topics outside their discipline. An objective of this work is to extend the ability of designers to solve more complex problems and thus navigate large concept sets outside their area of expertise. The second strategy involves taking the ten selected ideas from the entire set which scored the highest value of *Span* * *Detail* (concepts that are novel and well described) from the ML-based metrics. A third strategy consisted of selecting ten design concepts at random as a primary control (baseline). A fourth strategy consisted of ten sentences from an historical epic novel, i.e., Homer's Odyssey, and is meant to serve as a secondary control (worst case). Refer to Table 11 of Appendix A for a full listing of the ten concepts selected from each approach.

In order to evaluate performance, each combination of two of these 40 ideas was competitively ranked in a pairwise comparison. In an online crowdsourced platform, users were presented with two ideas in a random order for each human intelligence task or "HIT." Each permutation, or unique pairing of 40 by 40 concepts, was tested with three replicates. The average scores are reported in the format of a dominance matrix, typically used to determine rankings or performance of a team or strategy across many individual matches or comparisons [45]. Figure 5 shows the prompt.

Ontologically selected concepts were selected as more novel and detailed the most often, followed by human, random, and control, consecutively. The following differences are significant using a pairwise binomial comparison (transformed z-test of defective populations) for each order in the rank: ontological metrics > human ($p = 0.06$), nearly significant; human > random ($p = 0.01$), strongly significant; random > odyssey ($p < 0.001$), strongly significant. The full sets of concepts that were compared are provided in the Appendix. Figure 6 reports the dominance test results. Interestingly, some passages from the novel, which is in fact about

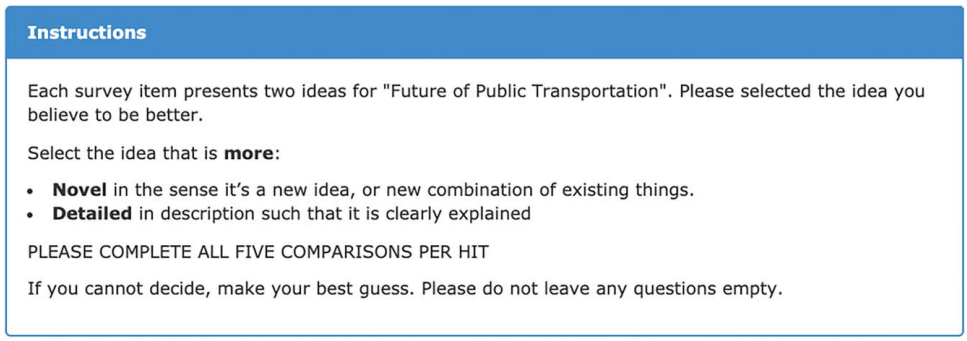


Fig. 5 Instructions provided for binary Mturk [21,22] comparisons

	ODYSSEY	RANDOM	HUMAN	ML METRICS	Row Avg. % Winning
ODYSSEY	49%	34%	22%	17%	30%
RANDOM	66%	50%	24%	24%	41%
HUMAN	77%	75%	49%	42%	61%
ML METRICS	82%	75%	57%	50%	66%
Column Avg. % Losing	69%	59%	38%	33%	

Fig. 6 Dominance matrix results reporting average scores for Mturk participant responses to pairwise comparisons of design concepts. See Fig. 5 for the prompt and Appendix A for the list of design concepts. A total of 4800 comparisons were made.

Downloaded from http://asmedigitalcollection.asme.org/mechanicaldesign/article-pdf/142/3/031113/6649246/md_142_3_031113.pdf?casa_token=JvwSKaiw1oIAAAAA:HtUkRLGCuCc63WS_0PHGIPVY4zKH5xZbvnFwU8ToUCo_w9Zyd3119WtAa2ZSjYLxf02k2w by Universita Di Pisa user on 25 Novembe

Table 9 Historical evidence of sample designs at two time points, as explored by Google search results first page

Using a log scale for quartile segmentation	*Novelty* * *Detail* range	Quartile population	*n* samples	*n* (%), seen by January 2016	*n* (%), seen by April 2019
First quartile	[0.1–1.0]	101	25	17 (68%)	18 (72%)
Second quartile	(1.0–10]	481	120	78 (65%)	84 (70%)
Third quartile	(10–100]	381	95	52 (55%)	63 (66%)
Fourth quartile	(100–1000]	21	6	1 (17%)	1 (17%)

Note: Two hundred and forty six evenly spaced samples were taken (one sample in every four data points). Note that less highly scored (less novel designs) are more likely to have already been developed.

exploration, do obliquely refer to navigation and were selected in a small percent of trials.

4.3 Predicting Technology Trends: Industrial Emergence. A third application for ML metrics is to provide, alongside crowdsourced ideation, a predictive capacity for technologies and products that are likely to emerge. The results of the distributed ideation exercise were collected on approximately three years before this study. As some time has elapsed in developing the toolkit requisite to successfully compete against human selection processes, a unique opportunity emerged, which is to explore whether any of the highly ranked concepts have since been deployed in industry or society.

This is a pilot test of the predictive capability of this overall approach. To evaluate whether the *Span* * *Detail* metric is a predictor of industrial emergence, the results are examined to determine whether the more novel solutions were less likely to already exist.

4.3.1 Raw Metric for Selection. Samples are taken at regular intervals across the entire data set. The sampling rate one in every four concepts is expected to provide 95% confidence with approximately 5% margin of error based on sample size power calculation. The authors evaluated, from a set of evenly distributed sample concepts, whether there was evidence as per Google search of a given transportation solution on or before two dates of interest. If a relatively new technology was identified, deeper search was conducted to determine when it was announced or the company which produces that technology was founded. The results of this sampling test can be seen in Table 9. There were fewer technologies that had already been developed in the high novelty categories. However, this changed over time with a significant growth in the third quartile. This is a positive indication of a potential correlation between the ML-metric-based assessment and industrial emergence.

The distribution of *Span* * *Detail* scores follows an inverse proportional morphology, consistent with Zipf's law [46]. Therefore, to analyze the results, the samples are grouped into four clusters or "quartiles" based on a logarithmic scale (e.g., 0.1 to 1, 1 to 10, and so on). The expectation is that lower quartiles would have fewer existing concepts because these are predicted to be the less novel regions. Some low scoring solutions were relatively ancient technology such as "Use a Horse and Buggy" which appeared more than once in the first quartile sampling. Some high scoring concepts are yet to be implemented, many of which appeared in the fourth quartile sampling, e.g., "I'm envisioning a system of small feet-sized platforms, along with a handle at arm-height, running on street-level light rail tracks throughout the city. Like an escalator, these continuously run in a loop around the city/neighborhood, and passengers can jump on and off at will. (Access would be prepaid and mediated by a radio-frequency identification (RFID) chip card and scanner on the devices.) These would probably move at about 5–10 miles per hour—faster than regular walking, but slower than other forms of transport that require more infrastructure and potential for danger…." Pairwise transformed z-test of defective populations comparisons are used to determine significance among the ratio of historically emerged technology. If less novel (common) solutions have existed for longer by definition, then it stands to reason that novel solutions were either recently developed or are yet to be developed. Statistical significances (comparison of proportions) are as follows for whether the solution is significantly less likely to have occurred in either 2016 or by 2019 pairwise by quartile: 4th > 3rd ($p < 0.001$) which is statistically significant; 3rd > 2nd ($p = 0.067$); 2nd > 1st ($p = 0.36$).

4.3.2 Core Entity for Selection. In some cases, not all regions in a design space may be equally valid. One way to further guide selection in a more targeted manner is to use the *Core Entity* as a mechanism to focus on a specific area. For illustration, the following hypothetical case is proposed "for a particular project a design team is tasked to redesign public bus systems to be more efficient," then those concepts are identified that have "bus" tagged as a *Core Entity*. Next, a supplementary pilot study was conducted to take a closer look at the two top scoring concepts with the *Core Entity*, "bus," cluster to search for innovative bus-related technologies. The highest scoring concept, that was also tagged with the *Core Entity*, "bus" was a form of social network for ridesharing network based on the rider's workplace. A pilot of this concept had been launched by MIT in 2011 [47].

The second highest scoring concept is as follows:

> *…the trolley or bus could be made to target things people want to go do instead of just transportation. Like grocery shopping or night out to watch movie theatre and that way everyone is grouped and the businesses upsales, etc. can develop better because of that.* (Crowdsourcing Participant, 16 Jan. 2016)

To the knowledge of the authors, there was no on-demand bus service, at least in Singapore (where the study was conducted) at the time initial concepts were collected, in 2016. However, the local transit authority recently announced the deployment of a pilot project to evaluate an on-demand public transportation system consisting of an app providing adaptive on-demand bus routing which would cater to exactly this concept (see Fig. 7). The pilot program was announced for deployment on December 17, 2018, approximately three years after this entry was submitted for ideation [48].

5 Discussion

In this paper, a method is presented for automatically scoring design concepts through ML-based creativity metrics from a large set of crowdsourced design concepts using a machine learning-based approach. Ontological properties are extracted from concepts using a machine learning package trained on the FreeBase knowledge graph, Wikipedia, and a variety of structured and unstructured data, using the Textrazor API, while Textrazor includes the required functions there are named entity recognition approaches and software available with comparable function. Metrics to support creativity assessment are proposed based on analysis of the outputs from the API. The method and equations for determining these metrics from outputs of the ontological analysis [23] are provided.

Design concepts were generated and evaluated from four different ideation cases to evaluate the method. In each ideation case, the saturation of unique *Core Entities* was analyzed using asymptotic

Downloaded from http://asmedigitalcollection.asme.org/mechanicaldesign/article-pdf/142/3/031113/6649246/md_142_3_031113.pdf?casa_token=JvwSKaiw1oIAAAAA:HtUkRLGCuC63WS_0PHGIPVY4zKH5xZbvmFwU8ToUCo_w9Zyd3119WtAa2ZSjYLxf02k2w by Universita Di Pisa user on 25 Novembe

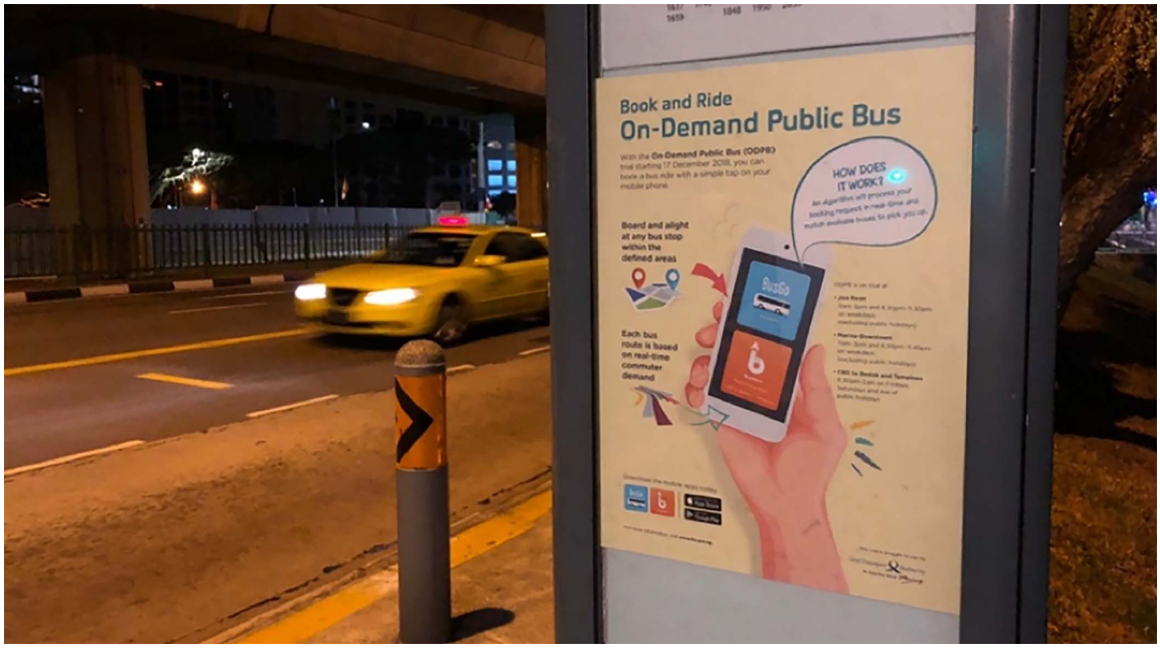


Fig. 7 Public service announcement for the recently deployed on-demand bus new service experimental pilot in Singapore, Singapore, trial initiated December 17, 2018

convergence analysis according to a procedure developed by Lim et al. [3]. Each case successfully converged at $n = 1000$ entries or less. This indicates that the design space coverage was sufficiently broad.

To emphasize the potential impactof an automated selection strategy and better understand the distribution of design scores, the results are plotted (Fig. 8). Regression models showed a strong correlation with an inverse proportional model, analogous to Zipf's law. In short, by integrating the area under the curve graph covered by the first ten ideas alone (for one problem, out of 1000 candidates), approximately 15% percent of the entire integral is provided by the ten highest ranked concepts though they account for only 1% of the total number of concepts. In short, there are a small number of concepts that scored very highly. This helps to provide one possible explanation why the ML-metric-based selection would score highly. Since there are relatively few high scoring designs, it would take the human rater significant time and concentration to identify the highest scoring concepts.

Several tests serve to evaluate the performance of these metrics to categorize and rate design concept sets. The first test evaluates the inter-rater agreement (Pearson's correlation) achieved between the ML metrics and expert assessment of novelty and level of detail. A second test determines which of four subsets of ten ideas are preferred through a crowdsourced binary dominance matrix trial series of over 3800 one-to-one comparisons. This is accomplished with replicated binary pairwise comparisons of each possible pair from the four subsets of ten ideas (see Table 11 of Appendix A). The four selection strategies were selection by a human with a time limit, selection of the highest scored ideas based on ML metrics, random selection, and control in the form of sentences from a

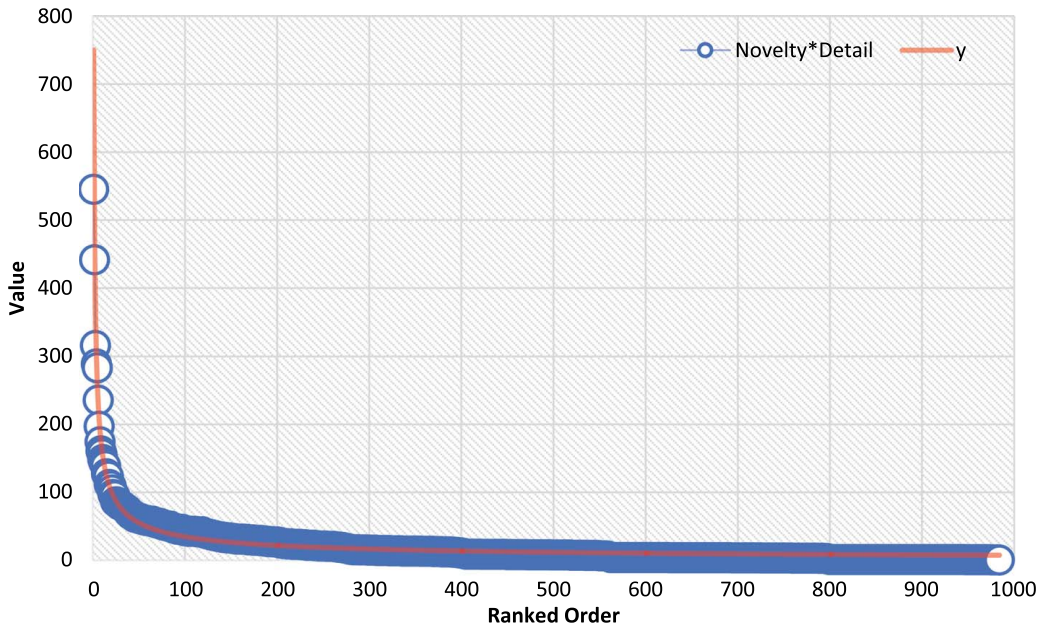


Fig. 8 Distribution of *Span*Detail* scores for problem case (2) from Table 1

oaded from http://asmedigitalcollection.asme.org/mechanicaldesign/article-pdf/142/3/031113/6649246/md_142_3_031113.pdf?casa_token=JvwSKaiw1oIAAAAA:HtUkRLGCuCc63WS_0PHGIPVY4zKH5xZbvmFwU8ToUCo_w9Zyd3119WtAa2ZSjYLxf02k2w by Universita Di Pisa user on 25 Novembe

classical novel. The most dominant results were from the ML-metric selection strategy. The expert rating to ML-metric inter-rater agreement was performed on a subsample of ideas from each problem. Selection using the ML metrics was performed over the entire data set of over one thousand concepts for each design problem. Thus, it is unlikely that the findings were due to selective tuning to the test sample. A third test explored technology progression for the public transportation problem to show highest scored ideas for *Span* and *Detail* to determine first whether there is a correlation with novelty and historical existence and whether some of the novel technologies were developed after a waiting period. It was observed that there were significantly fewer design concepts in the highly novel segments that had been implemented. However, some of these were newly implemented in the three year gap which indicates that more novel scoring designs may serve to predict industry emergence.

Methods for graphical representation of the results could also be explored. One highly readable format for organization design concepts is a mind map. Mind maps are a popular design tool due to their hierarchical clustering which enables summarization of a design space visually [49,50]. Mind maps were selected based on previous research that mind mapping enhances organization and retrieval of data [49]. A mind map was created to visually represent the highest scoring ideas from one of the ideation cases as an exemplar (Appendix B). The highest scoring concepts from distributed ideation for public transport (*case-2* in Table 1) using the *Span* * *Detail* metric are summarized and presented in a mind map to illustrate the output of this method for potential use in ongoing design activities. Appendix B depicts a mind map of abbreviated highlights of the top concept. The *Core Entity* from each concept is used as a cluster label (see Fig. 9 of Appendix B).

6 Limitations

There are several avenues for future work of potential value to support diving deeper into automated creativity assessment with computational approaches. First, a broader array of metrics and their formulation could be empirically evaluated. Machine learning could be applied to evaluate numerous alternative formulations based on the ontological data available to induce a set of stronger design metrics. In the current work, the metrics were developed through a synthesis of literature review and empirical evaluation. Furthermore, the preliminary experiments in inter-rater agreement can be substantially expanded to support developing and refining the proposed metrics and to identify new metrics of value in automated concept ranking. The metrics and means of analysis provided by humans and ML are not identical. This could have several impacts on results. One such impact is that a well-described and useful solution could be "lost" in the intermediate scoring results. Generally, someone using these metrics would be encouraged to explore both high scoring concepts and concepts within several *Core Entities* of interest to increase the chances of finding a relevant concept.

Most design evaluation approaches require an explicit approach to evaluate feasibility or quality. The current approach only provides a guidance strategy. Although sorting by *Core Entity* may enable the designers to more rapidly identify concepts of interest, there is no guarantee that these concepts will be valid or highly feasible. Future research must continue to explore potential avenues to compute feasibility.

Qualitatively, in reviewing specific scores, one can see that the ML scored concepts are more "obvious" while concepts that scored highly for humans and low for ML may involve a subtle interpretation. There is a delicate balance between succinctness and inspiration. It is perhaps unclear how to evaluate the difference between an elegant trigger that inspires a good idea and a clear description of the good idea itself. Perhaps both are equally valuable. In any case, the ML tool would be less likely to value a "trigger" as highly as a well-described idea. Metaphorically, a "Picasso" would be poorly rated, while the "Rembrandt" would score highly. In some sense, the ML may be more objective overall, but this may have a cost of being ignorant toward certain forms of elegance.

There is a potential risk that ideas might be incorrectly over-rated as creative when using this method. This is because classification by neural networks is a probabilistic process. In the proposed method, this is mitigated by collecting a large data set and applying asymptotic saturation analysis. With more ideas to assess, it follows that there would be more highly creative ideas. This is supported by the score distribution seen in Fig. 8. In short, since the approach is probabilistic, it may be less effective on small data sets.

7 Conclusions

Overall, this work demonstrates one set of metrics and a method for automating evaluation of design concept sets using machine learning. Source data for measurement was from early stage ideation efforts in the form of design concept solution sets. Other applications could include standardized experimental evaluation in ideation study protocols; or application by designers working in practice to rapidly identify potentially fruitful avenues of exploration.

Designers might use the proposed approach in several different ways. For example, those ideas with the highest *Span* and *Detail* scores were dominant against human selected concepts for "high novelty" assessment. Thus, perhaps such a filtering could be used to identify long-term new product service and system development opportunities. *Core Entity*-based categories may serve as a predictor of feasibility, though this hypothesis is not formally tested herein. For example, the most frequent category was "bus" (frequency of occurrence of *Core Entity of* 6%). Many of these concepts are readily implementable. Less frequently identified *Core Entities* like "air bikes" (frequency of occurrence of *Core Entity of* 0. 1%) and "jetpacks" (frequency of occurrence of *Core Entity of* 0. 1%) were of the lowest frequency and could be argued to be less feasible, e.g., "jetpacks" have existed in research and development for decades but are yet to be implemented in mass transit.

The *Core Entity* tags can be used to target a subset of ideas within an area of interest. As an illustrative example, one could select the subset concepts with "bicycle" or "roadway" as the *Core Entity* and review these ideas in planning development of a new dedicated bike lane.

Core Entity tags can be used in combination with the ML metrics to automatically generate solution mind maps. Visualizations could also be useful in automatically analyzing user needs or extracting insights from other types of design data. Recently, NASA converted its lessons learned library structure to a mind map to support more meaningful search [51]. Study of cognitive science concepts such as attention, priming, or memory engrams could support the design of new automated ontological metrics.

Another avenue for ongoing and expanded research would be to go so far as to eliminate the need for crowdsourcing of concepts, or, perhaps, more practically, augment and complement the crowdsourced ideas. This would entail identifying an appropriate database and method for extracting key relevant content before applying the approach described above. One such example would be the BASE open source research database.[3]

The broader potential impact is to push automation into earlier phases of design, not to replace the human designer but to empower them to gain deeper insights from existing data.

Acknowledgment

This work is funded by the SUTD-MIT International Design Centre (IDC; Funder ID: 10.13039/501100010707) in Singapore.[4] Any opinions, results, or conclusions in the paper are those of the authors and do not necessarily reflect the views of the sponsor.

[3]https://www.base-search.net/about/en/index.php
[4]https://idc.sutd.edu.sg/

Downloaded from http://asmedigitalcollection.asme.org/mechanicaldesign/article-pdf/142/3/031113/6649246/md_142_3_031113.pdf?casa_token=JvwSKaiw1oIAAAAA:HtUkRLGCuCc63WS_0PHGIPVY4zKH5xZbvnFwU8ToUCo_w9Zyd3119WtAa2ZSjYLxf02k2w by Universita Di Pisa user on 25 Novembe

Appendix A

Table 10 Definition of key terms

Term, concept	Definition	Data structure (if applicable)
Design concept entry (entries)	A natural language-based (text) design concept sourced from an in-person (part of a group) or online participant	String of characters, "Sentence" form
Case-1	Design opportunity-problem for crowdsourcing design concepts: "Design future public transportation in your city"	n/a
Case-2	Design opportunity-problem for crowdsourcing design concepts: "Imagine a new transportation system in your neighborhood—Describe it!"	n/a
Case-3	Design opportunity-problem for crowdsourcing design concepts: "Explore possible inventions given a new technology: GPS that is accurate to less than 1 cm"	n/a
Case-4	Design opportunity-problem for crowdsourcing design concepts: "Explore possible inventions given a new technology: Weather prediction that is 100% accurate"	n/a
Entity (entities)	An entity is single unique real-world object, and entities are typically connected to single words or a word pair from a source text	String of characters, single word
Topic	A topic is frequently defined as "what the sentence is about;" topics occur at the sentence level	String of characters, single word
Category	A category is defined in relation to entities. Unique entities may have a hierarchical classification; in this case, a category is the superset to which a specific entity belongs	String of characters, single word
IPTC	IPTC is a standard for news media classification and is a broad and hierarchical knowledge classification system used in meta-tagging	String of characters, word or phrase
Edge	An edge is (together with vertices) one of the two basic units out of which graphs are constructed. Each edge has two (or in hypergraphs, more) vertices to which it is attached, called its endpoints	n/a
Vertex	A vertex (plural vertices) or node is the fundamental unit of which graphs are formed: an undirected graph consists of a set of vertices and a set of edges (unordered pairs of vertices), while a directed graph consists of a set of vertices and a set of arcs (ordered pairs of vertices). In a diagram of a graph, a vertex is usually represented by a circle with a label, and an edge is represented by a line or arrow extending from one vertex to another	n/a
Knowledge graph	A knowledge graph describes real-world entities and their interrelations organized in a graph, defines possible classes and relations of entities in a scheme, and allows for potentially interrelating entities with each other and covers various topical domains [52]	Graph
FreeBase	FreeBase is a specific knowledge graph project, within it, 39 million unique entities are connected based on some meaningful real-world relationship between those entities [53]	String of characters, word or phrase
Relevance, R_i	Relevance is a rational number on a scale of 0 to 1, with 1 being the most relevant. Relevance is determined by the contextual similarity between two entities in a knowledge graph [23], i.e., number of edges between them on a normalized scale	Rational number [0,1]
$R_{\text{core_entity}}$	Relevance of the Core Entity. The relevance of the Core Entity is the maximum relevance score of any entity in a given entry	Rational number [0,1]
Core Entity	Within a sample of text, the core entity is one which has the highest relevance to the entry topic	String of characters, single word
Lemma	A lemma is the canonical form, dictionary form, or citation form of a set of words (headword)	String of characters, word stem
Span	The sum of distances of each entity in the concept to the central named Entity, as measured by minimum spanning path within the knowledge graph	Rational mumber
Detail	A basic count of named entities within the concept	Rational number
ML metrics	These are design metrics calculated using outputs from the tagging process	Rational number
Saturation	Saturation is defined, typically in a context of data collection or testing, as where additional sample collection will no longer provide additional new knowledge	Number of entries at saturation
Fault tolerant	A fault tolerant system is one in which a particular subfunction may fail without affecting the overall system critical performance	n/a

oaded from http://asmedigitalcollection.asme.org/mechanicaldesign/article-pdf/142/3/031113/6649246/md_142_3_031113.pdf?casa_token=JvwSKaiw1oIAAAAA:HtUkRLGCuCc63WS_0PHGIPVY4zKH5xZbvmFwU8ToUCo_w9Zyd3i19WtAa2ZSjYLxf02k2w by Universita Di Pisa user on 25 Novembe

Table 11 Ten ideas/content provided in the comparative dominance matrix test

*ML metric (Span * Detail)*

A Bus system, within the metro area that was funded by a tax on gas purchased by individual commuters. In other words, free to the people who use the bus, and costs the people who don't. Have the busses running from a set time like 6am to 9 pm and pickups are every 15 min, so you never need to bother with knowing the bus schedule. Also, have all busses make a stop in the centre of town so if you need to get across town, take a bus from you house to the centre, and from there to your final destination. There should be a bus passing every school, hospital, shopping centre, library etc. Simply run smaller buses on less popular routes, use smaller buses in general. make one special route for city to city transfers. The system needs to be free or very affordable, fast and convenient. No need to waste resources and infrastructure on schedules, call centres, web pages, just keep it simple and convenient.

Imagine a small grass field, similar to the ones you find at playgrounds, with a small parking lot. Within this parking lot would be a couple of electric powered cars attached to power stations. These setups would be available every few blocks (about as frequent as modern bus-stops). Within the field beside the cars stations would be stationary work out equipment. These would be modelled after the free-to-use exercise stations in places like China. They would have equipment such as: bench presses, ellipticals, bikes, etc. All of these would be wired underground to the charging station attached to the cars. The energy created from the movement of each device will directly charge the cars in the station. Off to the side of each piece of equipment would be a touchscreen. A person can set up an account at their local government buildings, providing a valid driver's license, e-mail address, and credit or debit card number. When signed in, each account can track how much a person exercised and convert it to an 'available miles for usage' number. This means that the more a person uses this equipment, creating more energy for the cars, the more miles the person would be allowed to use the car for. This however also means, that prior to using the car they would have to deduct the amount of miles they plan to use from their account. If the person should go over their allotted miles, they will then be subsequently charged on their provided credit card. This encourages people to work-out more at these stations, so they can avoid the fees for using the car. This set-up would immensely help those who financially can't afford their own form of transportation, while promoting a healthier lifestyle in a way that could easily fit into our society's demanding everyday regime.

…app on my phone to type a location of the place i need to go. This would in turn contact all local…systems of…transport…irrespective of the vendors available or government supported transport systems which include Cabs, Buses, Trains, Subway…

An automated system of cabs. Not cars though, more like carriages driven by artificial intelligence (AI) "Horses" (Imagine that new Harley electric motorcycle engine, in a short, small, bullet shaped car like you used to see on the SIDE of a motorcycle, pulling a carriage, with an AI brain controlling it). A system like this would have the novelty and functionality of pulling people when you wanted to, or you could just use the AI 'sidecar' to run courier errands on the ground for companies that wanted fast delivery. Think like Amazon drones without the flying, the noise or the whole 'being small enough to get shot down by a .22 and stolen' thing. It keeps the battery away from the passengers if there is ever a crash, too. Which is good because those things can explode sometimes. The best thing about it, though? When it's just moving small items around for parcel delivery, it's not wasting fuel hauling an empty cab around with it. I can't stand seeing people waste so much energy driving a giant sport utility vehicle (SUV) around with no passengers or cargo. That act alone should be a top priority for politicians pushing for environmental reform to be talking about.

I live in a small desert city with a major traffic congestion issue. However, despite the grid layout of the town, there is a diagonal irrigation canal running from the northwest corner to the southeast corner. Currently it has some poorly maintained walking paths, but that's about the extent of its development. Now, in the desert, loss of irrigation water to evaporation under the glaring sun is an issue. I'd propose creating a covered walkway/bike path where solar panels provide protection to the water, as well as pedestrians and cyclists. We could even use that solar energy to create small electric scooter rental/charging booths. It would provide a shortcut - a green highway from one end of town to the other.

I'm envisioning a system of small feet-sized platforms, along with a handle at arm-height, running on street-level light rail tracks throughout the city. Like an escalator, these continuously run in a loop around the city/neighbourhood, and passengers can jump on and off at will. (Access would be prepaid and mediated by a RFID chip card and scanner on the devices.) These would probably move at about 5-10 miles per hour - faster than regular walking, but slower than other forms of transport that require more infrastructure and potential for danger. It'd kind of be like, Razor scooters but built into the city landscape and available at-will. Like a bike-share system, but always there and with no energy or exercise expended. This would be a great way for tired people to avoid walking, and would speed up trips to the closest subway station/bus stop.

Fully automated system of public transit; similar in function to a subway but smaller and without drivers (more like the automated tram systems used at airports). Probably using driverless car technology so it could utilize existing roadways? You would use a pass card to enter a specialized waiting area, like a covered bus stop, at which the driverless vehicle/tram/etc would stop at and allow entry (only to those allowed into the station; maybe functions like the turnstiles in a subway?). Vehicle would be the size of a small city bus but could connect to some city-run data network to find and avoid traffic stops and slowdowns (and use/switch between already existing bus lanes and subway tracks?). It could have its own roadway/railway too but it would be nice to not have to create something new. It would be more reliable than a bus system, which is rarely on time where I live, but more accessible than a subway/railway, which does not reach enough areas of my city for me to use frequently.

my neighbourhood future transportation system will be made up of flying cars. These cars will be able to take off and land like helicopters. There will be no more traffic jams, traffic light, annoying speed limits and harassing cops. People will be able to travel to any part of the world without going through airports and flight connections. There will be new telecommunication and navigation technology that will help the drivers navigate safely without having mid-air collisions and also mid-air gas stations to refuel the cars.

This is a hybrid airship. Technology has developed that allows these vehicles to be more compact and use hydrogen lift cells safely. There are dozens of air ferries lifting people quietly into the air and to major cities all across America. The carbon footprint of the consumer aviation industry is drastically reduced. The noise pollution of traditional airports is all but gone. The hybrid airship is the vehicle of choice in terms of transportation.

This idea would have to be built in a city near a river, or attached to a solar farm because it relies on using naturally produced hydroelectric or solar power. Basically, it's a train on an electrified track, that moves slowly and stops at intervals in a big loop throughout a city. It would make stops the same way buses make stops. If you rode it long enough, you'd wind up at the same stop you got on at. It relies completely on green energy, no exhaust just some electric motors driving the cars. People would be able to get on and just ride to where they needed to get off, leaving their personal vehicle in a parking area. It could be configured to have small, individual cars maybe fitted with a GPS system so you could tell it where you needed to go and it would get you as close as possible. The main idea is it would be free and the city could tax businesses a small amount for serving their parts of town, just to help pay maintenance costs.

HUMAN

I imagine a taxi speed boat that could take you downtown in a matter of minutes. It will help skip all the traffic on the street and cut your time in half. Perfect for people who live by bodies of water and are in a rush like NYC.

I thought of a light rail and below-ground subway system could be transport people to and from a downtown arena we have here in metro Columbus, Ohio.

oaded from http://asmedigitalcollection.asme.org/mechanicaldesign/article-pdf/142/3/031113/6649246/md_142_3_031113.pdf?casa_token=JvwSKaiw1oIAAAAA:HtUkRLGCuCc63WS_0PHGIPVY4zKH5xZbvnFwU8ToUCo_w9Zyd3119WtAa2ZSjYLxf02k2w by Universita Di Pisa user on 25 Novembe

Right now most people use cars and a few people use buses, but that's about it. Congestion could be decreased substantially provided people who normally commute by themselves to and from hockey games that take place two to three times a week. This would solve congestion issues on the nights where these games take place and reduce the need for a large amount of safety and security officers to direct traffic and play crowd control on weekends and Fridays.

From small towns to big cities, public transportation is helping to generate new vitality like never before. Beautiful, walkable town squares combine residential, retail, office, recreation and public transportation features to make community living easy. New bus services and light rail extensions improve mobility and ease congestion. Even the smallest adjustments, like widening sidewalks and adding bikeways, can provide dramatic impact and value.

Fully automated system of public transit; similar in function to a subway but smaller and without drivers (more like the automated tram systems used at airports). Probably using driverless car technology so it could utilize existing roadways? You would use a pass card to enter a specialized waiting area, like a covered bus stop, at which the driverless vehicle/tram/etc would stop at and allow entry (only to those allowed into the station; maybe functions like the turnstiles in a subway?). Vehicle would be the size of a small city bus but could connect to some city-run data network to find and avoid traffic stops and slowdowns (and use/switch between already existing bus lanes and subway tracks?). It could have its own roadway/railway too but it would be nice to not have to create something new. It would be more reliable than a bus system, which is rarely on time where I live, but more accessible than a subway/railway, which does not reach enough areas of my city for me to use frequently.

There are solar powered buses in my neighbourhood which are 100% sustainable and operate autonomously, very much like the commuter trams at some airports such as Miami International Airport. These require no labour, and as such, have no human element involved in decision-making. As they are programmed to stop at intervals (designated by where the bus stops would be) there is a set time to off-load and pick up customers. Of course, handicapped and large groups can press a certain button which would allow more time and/or ramps to deploy.

I would like to see an advanced transportation system that involved driverless cars. The idea is the same as using a taxi but without a driver. Like Uber, you would use your mobile device to order a car and control the experience. The car shows up at your door, takes you to your destination and you are billed for the ride. It would save me thousands of dollars in the cost to own, maintain, insure and license my own personal car.

I would like to see a modern underground railway system installed in my neighbourhood. It would be noiseless and very fast. The railway will connect all the neighbourhood together making it easier to move around. The train would be making stops every half hour to pick up and drop passengers.

This idea would have to be built in a city near a river, or attached to a solar farm because it relies on using naturally produced hydroelectric or solar power. Basically, it's a train on an electrified track, that moves slowly and stops at intervals in a big loop throughout a city. It would make stops the same way buses make stops. If you rode it long enough, you'd wind up at the same stop you got on at. It relies completely on green energy, no exhaust just some electric motors driving the cars. People would be able to get on and just ride to where they needed to get off, leaving their personal vehicle in a parking area. It could be configured to have small, individual cars maybe fitted with a GPS system so you could tell it where you needed to go and it would get you as close as possible. The main idea is it would be free and the city could tax businesses a small amount for serving their parts of town, just to help pay maintenance costs.

I imagine a new transportation system that was both friendly to the environment and fast. These new vehicles would be autonomous, run on electricity and maintained by the city and housed in a centralized location. When someone determines that they need to get to one place, they would simply summon the vehicle from their computer or a mobile phone and the vehicle would be there promptly to pick them up. In this city, there would be no other form of public transportation, no one would own their own vehicles or drive themselves and all vehicles' locations and speed are tracked through a central computer so that each "pod" or vehicle would be able to take the most efficient routes at the most efficient speeds.

I think a good idea will be to use electric cycles instead of a bike or a car in metros. They not only help in decreasing pollution as well as motivating people towards physical exercises. A proper network of e cycles and connected to a main frame can be seen as a new transportation system emerging. Moreover this can also be used as a public transportation system by constructing cycle stands where they can be charged and the cost of public transport can be paid in the machine.

RANDOM

An expansion of the sky rail or a similar system with connection to the sky rail could be added to the surrounding cities of Seattle. The addition of public transportation would cut pollution and reduce traffic congestion with the plus side of convenience. It would be a good idea to implement this idea while we still have space available, rather than wait until the problem is a burden.

A human-sized slot-car system. Where cars run independently self-driving down a track through electricity like a trolley car, but just big enough for a regular sized family (in other words size of a regular car). In order to move faster cars can link up by matching speed and then essentially "plugging" into the car ahead of it. This could go much faster than regular cars and not have any of the problems associated with regular self-drive cars seeing as they move on a track.

Moving walkways or sidewalks, users would just step on where they please and get off whenever they arrive at their destination. There could be faster lanes for those traveling further.

I think there should be bikes that you can easily rent. You can get them in the major shopping areas and drive them around town to get exercise as well as helping to use less gasoline.

Side walk tracks, like what you would see in an airport, would benefit local transit for elderly and disabled people in my neighborhood

There could be some kind of light rail system for the people. It would make stops at malls, universities, grocery stores, and various event centres.

Hydrogenase would be an aerial ship for a future low on oil. It€™s a hybrid airship that would be entirely reliant on biofuels for transport and that could function as a vessel for a variety of reasons.

Automated driverless cars networked to navigate together and accept passengers via mobile applications

A new transportation method could be using self-driving vehicles. During times when you are needing to pre drink before going to a bar to save money, you find yourself unable to drive to the said bar. With self-driving vehicles, you can let the robot do the critical thinking for you to ensure your safe arrival to your friendly neighbourhood bar. This would be fun and convenient. This would also reduce the amount of wrecks on a Friday and Saturday night.

My idea is a car application that notifies you when you get in about the weather and road conditions.

ODYSSEY

They reached the low lying city of Lacedaemon them where they drove straight to the of abode Menelaus [and found him in his own house, feasting with his many clansmen in honour of the wedding of his son, and also of his daughter, whom he was marrying to the son of that valiant warrior Achilles. He had given his consent and promised her to him while he was still at Troy, and now the gods were bringing the marriage about; so he was sending her with chariots and horses to the city of the Myrmidons over whom Achilles' son was reigning. For his only son he had found a bride from Sparta, daughter of Alector. This son,

oaded from http://asmedigitalcollection.asme.org/mechanicaldesign/article-pdf/142/3/031113/6649246/md_142_3_031113.pdf?casa_token=JvwSKaiw1oIAAAAA:HtUkRLGCuCc63WS_0PHGIPVY4zKH5xZbvnFwU8ToUCo_w9Zyd3119WtAa2ZSjYLxf02k2w by Universita Di Pisa user on 25 Novembe

Table 11 *Continued*

Megapenthes, was born to him of a bondwoman, for heaven vouchsafed Helen no more children after she had borne Hermione, who was fair as golden Venus herself.

'Let me tell you,' said I, 'whichever of the goddesses you may happen to be, that I am not staying here of my own accord, but must have offended the gods that live in heaven. Tell me, therefore, for the gods know everything. which of the immortals it is that is hindering me in this way, and tell me also how I may sail the sea so as to reach my home.'

'Stranger,' replied she, 'I will make it all quite clear to you. There is an old immortal who lives under the sea hereabouts and whose name is Proteus. He is an Egyptian, and people say he is my father; he is Neptune's head man and knows every inch of ground all over the bottom of the sea. If you can snare him and hold him tight, he will tell you about your voyage, what courses you are to take, and how you are to sail the sea so as to reach your home. He will also tell you, if you so will, all that has been going on at your house both good and bad, while you have been away on your long and dangerous journey.'

"And what, Telemachus, has led you to take this long sea voyage to Lacedaemon? Are you on public or private business? Tell me all about it."

"Bless my heart," replied Menelaus, "then I am receiving a visit from the son of a very dear friend, who suffered much hardship for my sake. I had always hoped to entertain him with most marked distinction when heaven had granted us a safe return from beyond the seas. I should have founded a city for him in Argos, and built him a house. I should have made him leave Ithaca with his goods, his son, and all his people, and should have sacked for them some one of the neighbouring cities that are subject to me. We should thus have seen one another continually, and nothing but death could have interrupted so close and happy an intercourse. I suppose, however, that heaven grudged us such great good fortune, for it has prevented the poor fellow from ever getting home at all."

"Do we know, Menelaus," said she, "the names of these strangers who have come to visit us? Shall I guess right or wrong?-but I cannot help saying what I think. Never yet have I seen either man or woman so like somebody else (indeed when I look at him I hardly know what to think) as this young man is like Telemachus, whom Ulysses left as a baby behind him, when you Achaeans went to Troy with battle in your hearts, on account of my most shameless self."

"How sad," exclaimed Telemachus, "that all this was of no avail to save him, nor yet his own iron courage. But now, sir, be pleased to send us all to bed, that we may lie down and enjoy the blessed boon of sleep.

I have come, sir replied Telemachus, "to see if you can tell me anything about my father. I am being eaten out of house and home; my fair estate is being wasted, and my house is full of miscreants who keep killing great numbers of my sheep and oxen, on the pretence of paying their addresses to my mother.

I was trying to come on here, but the gods detained me in Egypt, for my hecatombs had not given them full satisfaction, and the gods are very strict about having their dues.

Menelaus, son of Atreus, and you my good friends, sons of honourable men (which is as Jove wills, for he is the giver both of good and evil, and can do what he chooses), feast here as you will, and listen while I tell you a tale in season.

Appendix B

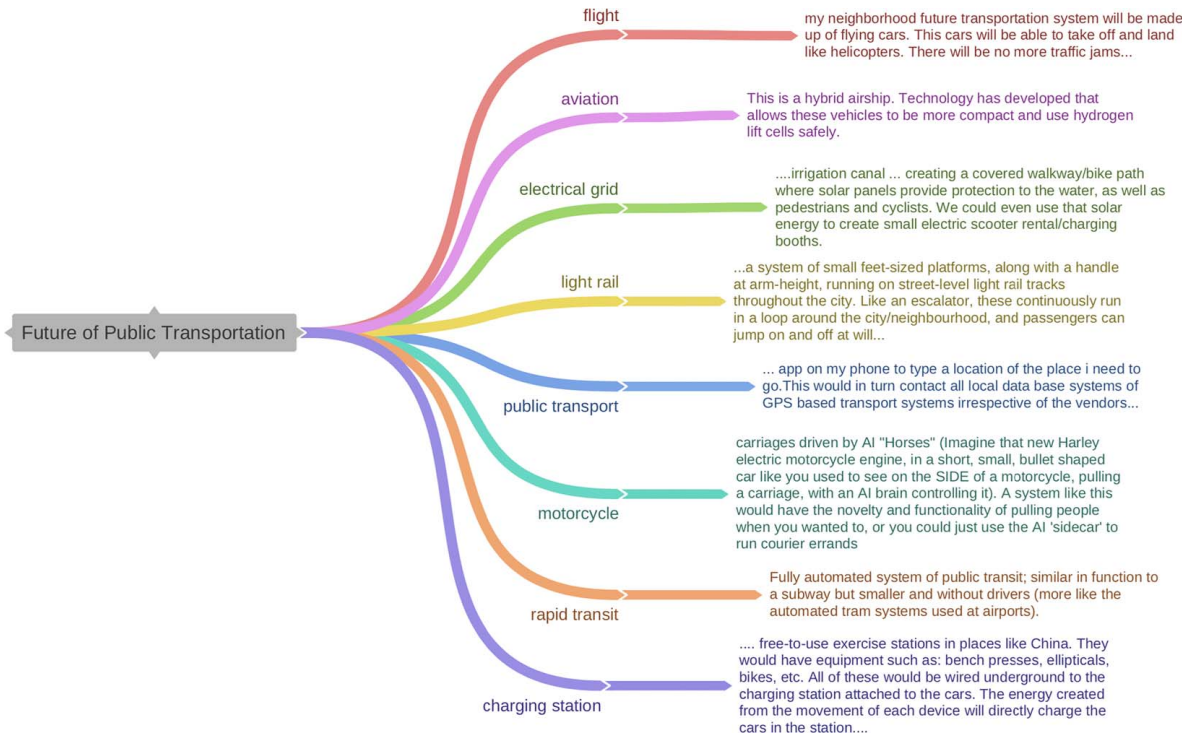


Fig. 9 Visual representation of automatically selected highly rated design concepts as a mind map. Developed by selecting the highest scoring ideas of the *Span* and *Detail* metrics. Note that these are the same concepts which were selected as most dominant; they have been abbreviated.

oaded from http://asmedigitalcollection.asme.org/mechanicaldesign/article-pdf/142/3/031113/6649246/md_142_3_031113.pdf?casa_token=JvwSKaiw1ojAAAAA:HtUkRLGCuCc63WS_0PHGlPVY4zKH5xZbvmFwU8ToUCo_w9Zyd3l19WtAa2ZSjYLxf02k2w by Universita Di Pisa user on 25 Novembe

References

[1] Kudrowitz, B. M., and Wallace, D., 2013, "Assessing the Quality of Ideas From Prolific, Early-Stage Product Ideation," J. Eng. Des., **24**(2), pp. 120–139.

[2] Linsey, J. S., Clauss, E. F., Kurtoglu, T., Murphy, J. T., Wood, K. L., and Markman, A. B., 2011, "An Experimental Study of Group Idea Generation Techniques: Understanding the Roles of Idea Representation and Viewing Methods," ASME J. Mech. Des., **133**(3), p. 31008.

[3] Lim, S. Y. C., Camburn, B. A., Moreno, D., Huang, Z., and Wood, K., 2016, "Design Concept Structures in Massive Group Ideation," ASME 2016 International Design Engineering Technical Conferences and Computers and Information in Engineering Conference, American Society of Mechanical Engineers, Charlotte, NC, Aug. 21–24, p. V007T06A006.

[4] Zhang, C., Kwon, Y. P., Kramer, J., Kim, E., and Agogino, A. M., 2017, "Deep Learning for Design in Concept Clustering," ASME 2017 International Design Engineering Technical Conferences and Computers and Information in Engineering Conference, Cleveland, OH, Aug. 6–9, American Society of Mechanical Engineers, p. V001T02A019.

[5] Song, B., and Luo, J., 2017, "Mining Patent Precedents for Data-Driven Design: The Case of Spherical Rolling Robots," ASME J. Mech. Des., **139**(11), p. 111420.

[6] Luo, J., Song, B., Blessing, L., and Wood, K., 2018, "Design Opportunity Conception Using the Total Technology Space Map," Artif. Intell. Eng. Des. Anal. Manuf., **32**(4), pp. 449–461.

[7] Shah, J. J., Smith, S. M., and Vargas-Hernandez, N., 2003, "Metrics for Measuring Ideation Effectiveness," Des. Stud., **24**(2), pp. 111–134.

[8] Fuge, M., Stroud, J., and Agogino, A., 2013, "Automatically Inferring Metrics for Design Creativity," ASME Paper No. DETC2013-12620.

[9] He, Y., and Luo, J., 2017, "The Novelty 'Sweet Spot'of Invention," Des. Sci., **3**, pp. 1–22.

[10] He, Y., Camburn, B., Luo, J., Yang, M. C., and Wood, K. L., 2019, "Visual Sensemaking of Massive Crowdsourced Data for Design Ideation," International Conference Engineering Design, Delft, The Netherlands, Aug. 5–8, pp. 409–418.

[11] Wortmann, T., and Tunçer, B., 2017, "Differentiating Parametric Design: Digital Workflows in Contemporary Architecture and Construction," Des. Stud., **52**, pp. 173–197.

[12] Wortmann, T., Costa, A., Nannicini, G., and Schroepfer, T., 2015, "Advantages of Surrogate Models for Architectural Design Optimization," Artif. Intell. Eng. Des. Anal. Manuf., **29**(4), pp. 471–481.

[13] Puentes, L., McComb, C., and Cagan, J., 2018, "A Two Tiered Grammatical Approach for Agent-Based Computational Design," Proceedings of the ASME 2018 IDETC 2018, ASME, Quebec City, Canada, Quebec City, Quebec, Canada, Aug. 26–29, pp. 1–11 .

[14] Campbell, M. I., Cagan, J., and Kotovsky, K., 1999, "A-Design: An Agent-Based Approach to Conceptual Design in a Dynamic Environment," Res. Eng. Des., **11**(3), pp. 172–192.

[15] Campbell, M. I., Cagan, J., and Kotovsky, K., 2000, "Agent-Based Synthesis of Electromechanical Design Configurations," ASME J. Mech. Des., **122**(1), pp. 61–69.

[16] Hauser, J., and Katz, G., 1998, "Metrics: You Are What You Measure!," Eur. Manage. J., **16**(5), pp. 517–528.

[17] Camburn, B. A., Auernhammer, J. M., Sng, K. H. E., Mignone, P. J., Arlitt, R. M., Perez, K. B., Huang, Z., Basnet, S., Blessing, L. T., and Wood, K. L., 2017, "Design Innovation: A Study of Integrated Practice," ASME 2017 International Design Engineering Technical Conferences and Computers and Information in Engineering Conference, Cleveland, OH, Aug. 6–9, American Society of Mechanical Engineers, p. V007T06A031.

[18] Shah, J. J., Kulkarni, S. V., and Vargas-Hernandez, N., 2000, "Evaluation of Idea Generation Methods for Conceptual Design: Effectiveness Metrics and Design of Experiments," ASME J. Mech. Des., **122**(4), pp. 377–384.

[19] Collado-Ruiz, D., and Ostad-Ahmad-Ghorabi, H., 2010, "Influence of Environmental Information on Creativity," Des. Stud., **31**(5), pp. 479–498.

[20] Oman, S. K., Tumer, I. Y., Wood, K., and Seepersad, C., 2013, "A Comparison of Creativity and Innovation Metrics and Sample Validation Through In-Class Design Projects," Res. Eng. Des., **24**(1), pp. 65–92.

[21] Paolacci, G., Chandler, J., and Ipeirotis, P. G., 2010, "Running Experiments on Amazon Mechanical Turk," Judgm. Decis. Mak., **5**(5), pp. 411–419.

[22] Kittur, A., Chi, E. H., and Suh, B., 2008, "Crowdsourcing User Studies With Mechanical Turk," Proceedings of the SIGCHI Conference on Human Factors in Computing Systems, Florence, Italy, Apr. 8–10, ACM, pp. 453–456.

[23] Crayston, T., "TextRazor: Technology" 2019, [Online], https://www.textrazor.com/technology, Accessed February 19, 2019.

[24] Derczynski, L., Maynard, D., Rizzo, G., Van Erp, M., Gorrell, G., Troncy, R., Petrak, J., and Bontcheva, K., 2015, "Analysis of Named Entity Recognition and Linking for Tweets," Inf. Process. Manage., **51**(2), pp. 32–49.

[25] IPTC, 2019, "List of IPTC NewsCodes and Other Vocabularies" [Online], http://cv.iptc.org/newscodes/.

[26] Troncy, R., 2008, "Bringing the IPTC News Architecture Into the Semantic Web," International Semantic Web Conference, Karlsruhe, Germany, Oct. 26–30, Springer, pp. 483–498.

[27] Becher, O., Hollink, L., and Elliott, D., 2017, "Exploring Concept Representations for Concept Drift Detection," SEMANTICS Workshops, Amsterdam, Sept. 11–14.

[28] Jean-Louis, L., Zouaq, A., Gagnon, M., and Ensan, F., 2014, "An Assessment of Online Semantic Annotators for the Keyword Extraction Task," Pacific Rim International Conference on Artificial Intelligence, Gold Coast, Australia, Dec. 1–5, Springer, pp. 548–560.

[29] Hollink, L., Bedjeti, A., van Harmelen, M., and Elliott, D., 2016, "A Corpus of Images and Text in Online News," Portorož, Slovenia, May 23–28, LREC.

[30] Linsey, J. S., Wood, K. L., and Markman, A. B., 2008, "Modality and Representation in Analogy," Artif. Intell. Eng. Des. Anal. Manuf., **22**(2), pp. 85–100.

[31] Fu, K., Chan, J., Cagan, J., Kotovsky, K., Schunn, C., and Wood, K., 2013, "The Meaning of 'Near' and 'Far': The Impact of Structuring Design Databases and the Effect of Distance of Analogy on Design Output," ASME J. Mech. Des., **135**(2), p. 21007.

[32] Gael, A. K., 1997, "Design, Analogy, and Creativity," IEEE Expert, **12**(3), pp. 62–70.

[33] Pellissier Tanon, T., Vrandečić, D., Schaffert, S., Steiner, T., and Pintscher, L., 2016, "From Freebase to Wikidata: The Great Migration," Proceedings of the 25th International Conference on World Wide Web, International World Wide Web Conferences Steering Committee, pp. 1419–1428.

[34] Google, "Data Dumps: Freebase Triples" [Online], https://developers.google.com/freebase/#freebase-rdf-dumps, Accessed April 28, 2018.

[35] Goldberg, Y., and Levy, O., 2014, "Word2vec Explained: Deriving Mikolov et al.'s Negative-Sampling Word-Embedding Method," preprint arXiv1402.3722.

[36] Goy, A., Magro, D., Petrone, G., Picardi, C., Rovera, M., and Segnan, M., 2017, "An Integrated Support to Collaborative Semantic Annotation," Adv. Human-Comput. Interact., **2017**, pp. 1–12.

[37] Eloranta, V.-P., Koskinen, J., Leppnen, M., and Reijonen, V., 2014, *Designing Distributed Control Systems: A Pattern Language Approach*, Wiley Publishing, New York.

[38] Summers, J. D., and Patel, A., 2018, "Supporting Design Decisions Using Data Driven Prediction Systems," Design Science Research (DSR) 2018: Workshop on Data Driven Design and Learning, Montreal, Canada, Aug. 23–25.

[39] De Nies, T., Beecks, C., Godin, F., De Neve, W., Stepien, G., Arndt, D., De Vocht, L., Verborgh, R., Seidl, T., and Mannens, E., 2016, "A Distance-Based Approach for Semantic Dissimilarity in Knowledge Graphs," 2016 IEEE Tenth International Conference on Semantic Computing (ICSC), Laguna Hills, CA, Feb. 3–5, IEEE, pp. 254–257.

[40] Bollacker, K., Evans, C., Paritosh, P., Sturge, T., and Taylor, J., 2008, "Freebase: A Collaboratively Created Graph Database for Structuring Human Knowledge," Proceedings of the 2008 ACM SIGMOD International Conference on Management of Data, Washington, DC, Dec. 15–19, ACM, pp. 1247–1250.

[41] Atilola, O., Tomko, M., and Linsey, J. S., 2016, "The Effects of Representation on Idea Generation and Design Fixation: A Study Comparing Sketches and Function Trees," Des. Stud., **42**, pp. 110–136.

[42] Tiong, E., Seow, O., Camburn, B., Teo, K., Silva, A., Wood, K. L., Jensen, D. D., and Yang, M. C., 2019, "The Economies and Dimensionality of Design Prototyping: Value, Time, Cost, and Fidelity," ASME J. Mech. Des., **141**(3), p. 31105.

[43] Saaty, T. L., and Ozdemir, M. S., 2003, "Why the Magic Number Seven Plus or Minus Two," Math. Comput. Model., **38**(3–4), pp. 233–244.

[44] Mark, G., Gonzalez, V. M., and Harris, J., 2005, "No Task Left Behind?: Examining the Nature of Fragmented Work," Proceedings of the SIGCHI Conference on Human Factors in Computing Systems, Portland, OR, Apr. 2–7, ACM, pp. 321–330.

[45] De Vries, H., Stevens, J. M. G., and Vervaecke, H., 2006, "Measuring and Testing the Steepness of Dominance Hierarchies," Anim. Behav., **71**(3), pp. 585–592.

[46] Newman, M. E. J., 2005, "Power Laws, Pareto Distributions and Zipf's Law," Contemp. Phys., **46**(5), pp. 323–351.

[47] Davis, R. T., 2011, "MIT Launches Social Network for Ridesharing," MIT News [Online], http://news.mit.edu/2011/mit-launches-zimride, Accessed February 25, 2019.

[48] Tan, C., "On-Demand Public Bus Services From Dec 17 as Part of 6-Month Trial," 2019, Straights Times [Online], https://www.straitstimes.com/singapore/transport/on-demand-public-bus-services-from-dec-17-as-part-of-6-month-trial, Accessed February 19, 2019.

[49] Buzan, T., 2005, *Mind Map: The Ultimate Thinking Tool*, Thorsons, London.

[50] Davies, M., 2011, "Concept Mapping, Mind Mapping and Argument Mapping: What Are the Differences and Do They Matter?," High. Educ., **62**(3), pp. 279–301.

[51] Meza, D., 2017, "How NASA Finds Critical Data Through a Knowledge Graph" [Online], https://neo4j.com/blog/nasa-critical-data-knowledge-graph/, Accessed February 19, 2019.

[52] Ehrlinger, L., and Wöß, W., 2016, "Towards a Definition of Knowledge Graphs," Semant (Posters, Demos, SuCCESS), 48.

[53] Pellissier Tanon, T., Vrandečić, D., Schaffert, S., Steiner, T., Pintscher, L., Bollacker, K., Evans, C., Paritosh, P., Sturge, T., and Taylor, J., 2008, "Freebase: A Collaboratively Created Graph Database for Structuring Human Knowledge," Proceedings of the 2008 ACM SIGMOD International Conference on Management of Data, Vancouver, Canada, June 9–12, ACM, pp. 1419–1428.

[54] Zurada, J. M., 1992, *Introduction to Artificial Neural Systems*, West Publishing Company, St. Paul, MN.

[55] Lafferty, J., McCallum, A., and Pereira, F. C. N., 2001, "Conditional Random Fields: Probabilistic Models for Segmenting and Labeling Sequence Data," Departmental papers (CIS), University of Pennsylvania.

[56] Perez-Ortiz, J. A., and Forcada, M. L., 2001, "Part-of-Speech Tagging With Recurrent Neural Networks," Proceedings of the International Joint Conference on Neural Networks (IJCNN'01) (Cat. No. 01CH37222), Washington, DC, July 15–19, IEEE, pp. 1588–1592.

oaded from http://asmedigitalcollection.asme.org/mechanicaldesign/article-pdf/142/3/031113/6649246/md_142_3_031113.pdf?casa_token=JvwSKaiw1oIAAAAA:HtUkRLGCuC63WS_0PHGIPVY4zKH5xZbvnFwU8ToUCo_w9Zyd3119WtAa2ZSjYLxf02k2w by Universita Di Pisa user on 25 Novembe